Tangent Spaces

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1 The lead question: velocities

Question. On a manifold, what is the *velocity* of a curve at a point p? (Velocity, not speed: speed requires a metric, which comes later.) It is traditional to call the velocity a vector, and in our formalism it will be one, but we first erase everything we learned about velocities and rediscover them.

Definition 1 (Velocity of a curve)**.** Let $(M, \mathcal{O}, \mathcal{A})$ be a smooth manifold and $\gamma : \mathbb{R} \to M$ a curve that is at least C^1 with respect to the charts of $\mathcal{A}$, with $\gamma(\lambda_0) = p$. The *velocity of* γ *at* p is the linear map

$$v_{\gamma,p} : C^\infty(M) \xrightarrow{\sim} \mathbb{R}, \qquad v_{\gamma,p}(f) = (f \circ \gamma)'(\lambda_0).$$

Here $C^\infty(M)$ is the set of all smooth functions $f : M \to \mathbb{R}$, which is a vector space under pointwise operations:

$$(f + g)(q) = f(q) + g(q), \qquad (\lambda f)(q) = \lambda\, f(q),$$

using addition and multiplication in $\mathbb{R}$. The map $v_{\gamma,p}$ is linear (derivative of a sum and of a scalar multiple).

Intuition. A child running along γ through a world with some temperature function f feels the rate of change of temperature along the way. That number $v_{\gamma,p}(f)$ is the *directional derivative* of f along γ at p. It must be defined for *every* f. This is a small shift in philosophy compared with $\vec{v} \cdot \nabla f = v^i \partial_i f$: rather than saying "the components v^i are the vector", the vector is the linear map that takes f to a number. If one knows how it acts on every function, one can recover the components by letting it act on the coordinate functions.

2 The tangent space

Definition 2 (Tangent space)**.** For each $p \in M$ the *tangent space* is the set

$$T_pM = \{v_{\gamma,p} \mid \gamma \text{ a smooth curve through } p\}$$

of velocities of all curves through p.

Different curves can give the same element; a curve traversed at double parameter speed gives a different element (it scales).

Remark (Intrinsic versus extrinsic)*.* Nothing in this construction refers to an ambient space in which M is embedded (as one might picture a tangent plane touching a sphere in $\mathbb{R}^3$). There was a long debate in differential geometry whether to define things intrinsically or via an embedding; embedding theorems say that it does not matter. Intrinsic is more economical, since if M is all of the universe there is nothing around it to refer to. The manifold only assumes that pieces of M look like $\mathbb{R}^d$, whereas an embedding might need an ambient $\mathbb{R}^N$ with $N \gg d$.

2.1 T_pM is a vector space

Observation: T_pM can be made into a vector space. Only vectors at the *same* point p can be added. Since elements are maps $C^\infty(M) \to \mathbb{R}$, define

$$(v_{\gamma,p} + v_{\delta,p})(f) = v_{\gamma,p}(f) + v_{\delta,p}(f), \qquad (\alpha \cdot v_{\gamma,p})(f) = \alpha\, v_{\gamma,p}(f),$$

using $+$ and $\cdot$ in $\mathbb{R}$. The remaining task is *closure*: the results must again be velocities of curves through p, i.e. lie in T_pM. (The vector space axioms follow from those of $\mathbb{R}$.)

Scalar multiples. Given γ with $\gamma(\lambda_0) = p$ and $\alpha \in \mathbb{R}$, define

$$\tau(\lambda) = \gamma\big(\lambda_0 + \alpha(\lambda - \lambda_0)\big),$$

i.e. run through the same curve with α times the parameter speed. Then $\tau(\lambda_0) = p$ and by the chain rule

$$v_{\tau,p}(f) = (f \circ \tau)'(\lambda_0) = \alpha\,(f \circ \gamma)'(\lambda_0) = \alpha\, v_{\gamma,p}(f).$$

So $\alpha v_{\gamma,p} = v_{\tau,p} \in T_pM$.

Sums. Let $\gamma(\lambda_0) = p = \delta(\lambda_1)$. Choose a chart (U, x) with $p \in U$ (this exists by the definition of a manifold). For λ near 0 define

$$\sigma_x(\lambda) = x^{-1}\Big((x \circ \gamma)(\lambda_0 + \lambda) + (x \circ \delta)(\lambda_1 + \lambda) - (x \circ \gamma)(\lambda_0)\Big).$$

The additions take place in $\mathbb{R}^d$ (in the chart), which is why a chart is needed. Then $\sigma_x(0) = x^{-1}(x(p)) = p$, because $(x \circ \delta)(\lambda_1) = x(p)$. For any $f \in C^\infty(M)$, insert $x^{-1} \circ x = \mathrm{id}$:

$$v_{\sigma_x,p}(f) = \big((f \circ x^{-1}) \circ (x \circ \sigma_x)\big)'(0) = \sum_{i=1}^{d} \partial_i(f \circ x^{-1})\big(x(p)\big)\ (x^i \circ \sigma_x)'(0),$$

by the multidimensional chain rule. Since $(x^i \circ \sigma_x)'(0) = (x^i \circ \gamma)'(\lambda_0) + (x^i \circ \delta)'(\lambda_1)$ (the constant drops out), running the chain rule backwards gives

$$v_{\sigma_x,p}(f) = (f \circ \gamma)'(\lambda_0) + (f \circ \delta)'(\lambda_1) = v_{\gamma,p}(f) + v_{\delta,p}(f).$$

So $v_{\gamma,p} + v_{\delta,p} = v_{\sigma_x,p} \in T_pM$.

Remark. The curve σ_x depends on the chart x, but its velocity at p does not: the final result involves only γ and δ. Hence the use of a chart is harmless. It does *not* mean one can add curves: that would amount to adding position vectors, which do not exist (there is no addition on M). It works only for the velocities at a single point.

3 Components with respect to a chart

Let (U, x) be a chart and γ a curve in U with $\gamma(0) = p$. For $f \in C^\infty(M)$ insert $x^{-1} \circ x$ and use the chain rule:

$$v_{\gamma,p}(f) = \big((f \circ x^{-1}) \circ (x \circ \gamma)\big)'(0) = (x^i \circ \gamma)'(0)\ \partial_i(f \circ x^{-1})\big(x(p)\big) \qquad \text{(sum over } i\text{)}.$$

Two pieces of notation:

$$\left.\frac{\partial}{\partial x^i}\right|_p f := \partial_i(f \circ x^{-1})\big(x(p)\big), \qquad \dot\gamma^i_x := (x^i \circ \gamma)'(0).$$

Here $\partial/\partial x^i|_p$ is a map $C^\infty(M) \to \mathbb{R}$; the symbol looks like an ordinary partial derivative but must always be translated back to the right side (partial derivative with respect to the i-th slot of the function $f \circ x^{-1} : \mathbb{R}^d \to \mathbb{R}$, evaluated at $x(p)$). Then

$$v_{\gamma,p} = \dot{\gamma}^i_x \left.\frac{\partial}{\partial x^i}\right|_p .$$

So the *components* of the velocity in the chart x are the derivatives of the chart representative $x \circ \gamma$ of the curve, and the **chart-induced basis** elements are $\partial/\partial x^i|_p$. They depend on the chart: a different chart gives different components, as in the example of a bird flying on a circle (in polar coordinates the corresponding curve is a straight line, and the components of its velocity look quite different from those in Cartesian coordinates). Components should be drawn in the chart, not in the real world.

Theorem 1. *Let (U, x) be a chart of the smooth atlas and $p \in U$. Then*

$$\left.\frac{\partial}{\partial x^1}\right|_p , \ \cdots , \ \left.\frac{\partial}{\partial x^d}\right|_p$$

form a basis of T_pM. In particular $\dim T_pM = d = \dim M$.

Proof. Spanning was shown above: every $v_{\gamma,p}$ is a linear combination of them. *Linear independence:* suppose $\lambda^i \, \partial/\partial x^i|_p = 0$ as a map on $C^\infty(M)$. Apply it to the coordinate function $x^j : U \to \mathbb{R}$ (a smooth function on the chart domain):

$$0 = \lambda^i \left.\frac{\partial}{\partial x^i}\right|_p x^j = \lambda^i \, \partial_i\big(x^j \circ x^{-1}\big)\big(x(p)\big) = \lambda^i \delta^j{}_i = \lambda^j,$$

since $x^j \circ x^{-1}$ maps $(\alpha^1, \ldots, \alpha^d) \mapsto \alpha^j$. So all $\lambda^j = 0$. □

Remark. The chart-induced basis lives only over the chart domain U, not in general over all of M (there may be no global chart, and even then no global coordinate-induced basis field). The vector space dimension of T_pM and the topological manifold dimension are different concepts that happen to agree.

Terminology. Any $X \in T_pM$ is the velocity of some curve through p (by the definition of T_pM), and, since T_pM is a vector space, it can be expanded in any basis, e.g. the chart-induced one:

$$X = X^i_x \left.\frac{\partial}{\partial x^i}\right|_p .$$

The X^i_x are the components of X with respect to that basis. The basis need not come from a chart, but if a chart has already been chosen it is convenient to use it.

4 Change of vector components under a change of chart

Warning. Older literature says "a vector transforms as…". Precisely: the *components* transform; the vector itself, being defined independently of charts (the velocity of the bird in the real world), does not change.

Let (U, x) and (V, y) be overlapping charts and $p \in U \cap V$. For $X \in T_pM$,

$$X = X^i_x \left.\frac{\partial}{\partial x^i}\right|_p = X^j_y \left.\frac{\partial}{\partial y^j}\right|_p .$$

To relate the components, apply $\partial/\partial x^i|_p$ to f and insert the identity $y^{-1} \circ y$:

$$\left.\frac{\partial}{\partial x^i}\right|_p f = \partial_i\big((f \circ y^{-1}) \circ (y \circ x^{-1})\big)(x(p)).$$

Here $y \circ x^{-1}$ is the chart transition map. The multidimensional chain rule gives

$$= \partial_i(y^j \circ x^{-1})\big(x(p)\big)\ \partial_j(f \circ y^{-1})\big(y(p)\big) = \left.\frac{\partial y^j}{\partial x^i}\right|_p \left.\frac{\partial}{\partial y^j}\right|_p f,$$

where $\partial y^j/\partial x^i|_p := \partial/\partial x^i|_p\, y^j$ is the notation introduced above. Since f was arbitrary,

$$\left.\frac{\partial}{\partial x^i}\right|_p = \left.\frac{\partial y^j}{\partial x^i}\right|_p \left.\frac{\partial}{\partial y^j}\right|_p.$$

Inserting into $X = X^i_x \partial/\partial x^i|_p$ and comparing with the expansion in the y-basis (basis vectors are linearly independent):

$$\boxed{X^j_y = \left.\frac{\partial y^j}{\partial x^i}\right|_p X^i_x}$$

This is the formula found in textbooks. Even if the change of chart is wildly nonlinear, at the single point p it acts on components by a *constant* matrix (the Jacobian of the transition map at $x(p)$).

Remark (Special relativity). Minkowski "space" of events is not a vector space either. The Lorentz transformations act linearly on the *velocity vectors* in each (flat) tangent space; requiring that positions also transform linearly is an over-structuring. One can do special relativity in polar coordinates too: a nonlinear change of position coordinates induces linear maps on tangent vectors at each point.

5 The cotangent space and the gradient

Since T_pM is a vector space, we can consider its dual, the **cotangent space**

$$T^*_pM = (T_pM)^* = \mathrm{Hom}(T_pM, \mathbb{R}).$$

Definition 3 (Gradient)**.** For $f \in C^\infty(M)$ and $p \in M$ define $(\mathrm{d}f)_p : T_pM \to \mathbb{R}$ by

$$(\mathrm{d}f)_p(X) = X(f).$$

This is linear in X, so $(\mathrm{d}f)_p \in T^*_pM$. It is called the *gradient of f at p.*

No chart is needed to define the gradient. It is a covector, a $(0,1)$ tensor over T_pM, not a vector.

Components. With respect to the chart-induced basis,

$$(\mathrm{d}f)_j = (\mathrm{d}f)_p\left(\left.\frac{\partial}{\partial x^j}\right|_p\right) = \left.\frac{\partial}{\partial x^j}\right|_p f = \partial_j(f \circ x^{-1})\big(x(p)\big),$$

which one computes with undergraduate analysis, giving one number per j.

Theorem 2 (Chart-induced dual basis)**.** *Let (U, x) be a chart, so that the coordinate maps $x^a : U \to \mathbb{R}$ are smooth functions. Then*

$$(\mathrm{d}x^1)_p,\ \ldots,\ (\mathrm{d}x^d)_p$$

is a basis of T^*_pM, and it is the dual basis *of $\partial/\partial x^b|_p$:*

$$(\mathrm{d}x^a)_p\left(\left.\frac{\partial}{\partial x^b}\right|_p\right) = \left.\frac{\partial}{\partial x^b}\right|_p x^a = \delta^a{}_b.$$

Change of covector components. Write $\omega \in T_p^*M$ as $\omega = \omega_j^{(x)}(\mathrm{d}x^j)_p = \omega_i^{(y)}(\mathrm{d}y^i)_p$. Evaluating on the basis vector $\partial/\partial y^i|_p$ and using the relation between the bases above,

$$\omega_i^{(y)} = \left.\frac{\partial x^j}{\partial y^i}\right|_p \omega_j^{(x)}.$$

This involves the matrix $\partial x/\partial y$, which is the *inverse* of the matrix $\partial y/\partial x$ used for vector components. Covector components transform in the opposite (inverse) way to vector components.

Warning. If one calls the gradient a vector and also uses the multidimensional chain rule, one gets a contradiction. It is not deep differential geometry that makes the gradient a covector: it is just the chain rule of undergraduate analysis.

Summary

- The **velocity** of a curve γ at p is the linear map $v_{\gamma,p} : C^\infty(M) \to \mathbb{R}$, $f \mapsto (f \circ \gamma)'(\lambda_0)$ (the directional derivative).
- T_pM, the set of all such velocities, is a d-dimensional vector space (scaling the curve parameter gives scalar multiples; a chart-based construction gives sums, with chart-independent result).
- A chart x induces a basis $\partial/\partial x^i|_p$ of T_pM; components are $\dot\gamma^i = (x^i \circ \gamma)'(0)$.
- Vector components change as $X_y^j = (\partial y^j/\partial x^i)|_p X_x^i$; the vector itself is chart independent.
- $T_p^*M = (T_pM)^*$; the gradient $(\mathrm{d}f)_p$ is a covector, with dual basis $(\mathrm{d}x^a)_p$. Its components transform with the inverse matrix.